VIT VELLORE · PROJECT REVIEW

Domain Name System (DNS) Lookup Tool

A focused interface for making name resolution observable.

M. Sathwik Reddy · Reg. No: 24BDS0310
T. Nikhil · Reg. No: 24BAI0315

Submitted to Faculty Sreethar S.

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THE PROBLEM DNS SOLVES

Why DNS matters: names become reachable destinations

People use names. Networks route to addresses.



Readable names

DNS maps human-readable domain names to network addresses and other service information.



Distributed scale

It replaced dependence on a single, centrally distributed hosts file as the Internet grew.



Observable translation

A lookup tool makes this translation visible, testable, and easier to troubleshoot.

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DISTRIBUTED NAMESPACE

DNS is a hierarchy of delegated responsibility

Client

Originator of DNS queries

Recursive Resolver

Performs lookups and caches answers

Root

Referral to appropriate TLD

TLD

Referral to authoritative server



QUERY CONTRACT

What a lookup tool actually asks for

Three inputs produce a structured, inspectable result.

1	2	3
FQDN	Record type	Class
Fully qualified domain name	A, AAAA, or another requested type	Typically IN for Internet data

STRUCTURED RESULT

Answer · TTL · Authority · Response status

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REQUEST PATH

From browser request to DNS answer

01

Client checks caches

Local and resolver caches are checked first.

02

Resolver follows referrals

The recursive resolver queries through the hierarchy.

03

Authority answers

The authoritative server returns the requested record.

04

Answer returns and caches

The resolver sends the answer back and caches it by TTL.

Recursion is the resolver's responsibility; iteration is the step-by-step referral path it follows.



IMPLEMENTATION VIEW

Implementation: a focused lookup pipeline

01

Validate and normalize

Check the domain name and normalize the requested record type.

02

Send the query

Use a resolver library or the system DNS API.

03

Parse the response

Read answer, authority, additional, status, and TTL fields.

04

Format the result

Present a clear terminal or simple web-interface response.

Practical error handling: invalid name, timeout, NXDOMAIN, and no matching record should each produce a distinct, understandable message.

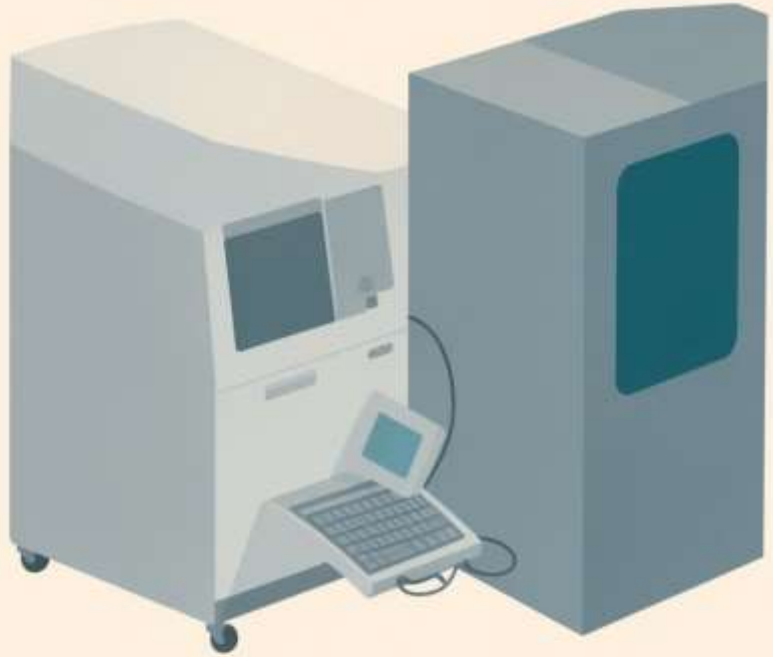
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LIVE DEMONSTRATION

Demonstration: inspect example.com

The exact returned values remain runtime output, not invented slide facts.

```
$ dns-lookup example.com --type A
Input domain name + record type
Runtime output appears here
Fields resolved value · record type · TTL · response status
$ dns-lookup example.com --type AAAA
Compare the IPv4 and IPv6 result sets
```



PRACTICAL VALUE

Where DNS lookup tools are useful



Diagnose reachability

Find misconfigured records behind a website that cannot be reached.



Verify changes

Check domain, hosting, and migration work as it propagates.



Inspect services

Review mail and service discovery records.



Teach networking

Turn requests and responses into observable learning material.

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TAKEAWAY

Make name resolution observable.

DNS turns a distributed naming system into a usable Internet service. A lookup tool exposes the inputs, path, answer, and failure mode behind a request.

NEXT STEP

Extend the prototype with **multiple resolvers**, **comparison views**, and **query history**.

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APPENDIX

References

NO.	REFERENCE	USE
01	Mockapetris, P. “Domain Names — Concepts and Facilities,” RFC 1034, RFC Editor.	DNS concepts
02	Microsoft Learn. “DNS Queries and Lookups in Windows and Windows Server”.	Query behavior
03	DNSLab. “Key RFCs — DNS Docs”.	RFC context

These sources support the hierarchy, query inputs, lookup behavior, and terminology used throughout the project review.

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